



VQBR: Variational Quantum Bayesian Regression via Measurement-Based MAP Direction Recovery

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Outline

- I. Introduction
- II. Proposed Variational Quantum Bayesian Regression
- III. Experimental Results
- IV. Conclusions and Future Work

I. Introduction

BAYESIAN LINEAR REGRESSION

- ❑ **Use prior information**

Combine observations with beliefs about regression weights.

- ❑ **Regularize the fit**

A Gaussian prior discourages weights that depart from the prior.

- ❑ **Recover a useful point estimate**

MAP provides one weight vector for regression and prediction.

I. Introduction

RESEARCH QUESTION

Classical reference

- ❑ Dense direct solve
 $O(ND^2 + D^3)$ arithmetic.
- ❑ Matrix-free CG and LSQR already
avoid explicitly forming $X^T X$.

Our question

- Can a near-term circuit estimate
the MAP objective through
measurable overlaps?

M. R. Hestenes and E. Stiefel, "Methods of conjugate gradients for solving linear systems," J. Res. Nat. Bur. Standards, vol. 49, no. 6, pp. 409–436, Dec. 1952.

C. C. Paige and M. A. Saunders, "LSQR: An algorithm for sparse linear equations and sparse least squares," ACM Trans. Math. Softw., vol. 8, no. 1, pp. 43–71, Mar. 1982.

I. Introduction

PROBLEM FORMULATION

- Data model (N samples, D features)

$$y = Xw + \eta, \quad \eta \sim \mathcal{N}(0, V \cdot I).$$

- Gaussian prior $w \sim \mathcal{N}(m_0, \Sigma_0)$.

- MAP estimation

$$w^* = \underset{w}{\operatorname{argmin}} \|Xw - y\|_2^2 + (w - m_0)^T M (w - m_0),$$

$$M = V\Sigma_0^{-1}.$$

A. N. Tikhonov and V. Y. Arsenin, Solutions of Ill-Posed Problems. Washington, DC, USA: V. H. Winston & Sons, 1977.

A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," Technometrics, vol. 42, no. 1, pp. 80–86, Feb. 2000.

II. Proposed Variational Quantum Bayesian Regression

SEPARATE SCALE FROM DIRECTION

$$w = t \cdot \phi(\theta), \quad \|\phi(\theta)\|_2 = 1.$$

- ❑ Direction $\phi(\theta)$: normalized real amplitudes selected by $U(\theta)$.
- ❑ Signed scale $t \in \mathbb{R}$: best point along the selected direction.

The circuit chooses a direction. A classical formula chooses the best signed scale.

II. Proposed Variational Quantum Bayesian Regression

SOLVE THE SCALE ANALYTICALLY

$$\mathcal{L}(t, \theta) = \alpha(\theta)t^2 - 2\beta(\theta)t + C.$$

$$\square \alpha = \phi^T (X^T X + M) \phi > 0.$$

$$\square \beta = \phi^T (X^T y + M m_0).$$

$$t^*(\theta) = \frac{\beta(\theta)}{\alpha(\theta)}.$$

$\alpha > 0$ under the positive-definite prior assumption.

For each fixed direction, scale optimization has an exact solution.

II. Proposed Variational Quantum Bayesian Regression

OPTIMIZE ONLY THE DIRECTION

$$\tilde{\mathcal{L}}(\theta) = -\frac{\beta^2(\theta)}{\alpha(\theta)} + C.$$

$$\min_{\theta, t} \mathcal{L}(t, \theta) = \min_{\theta} \tilde{\mathcal{L}}(\theta).$$

Guaranteed

Exact scale elimination
within the ansatz.

Still required

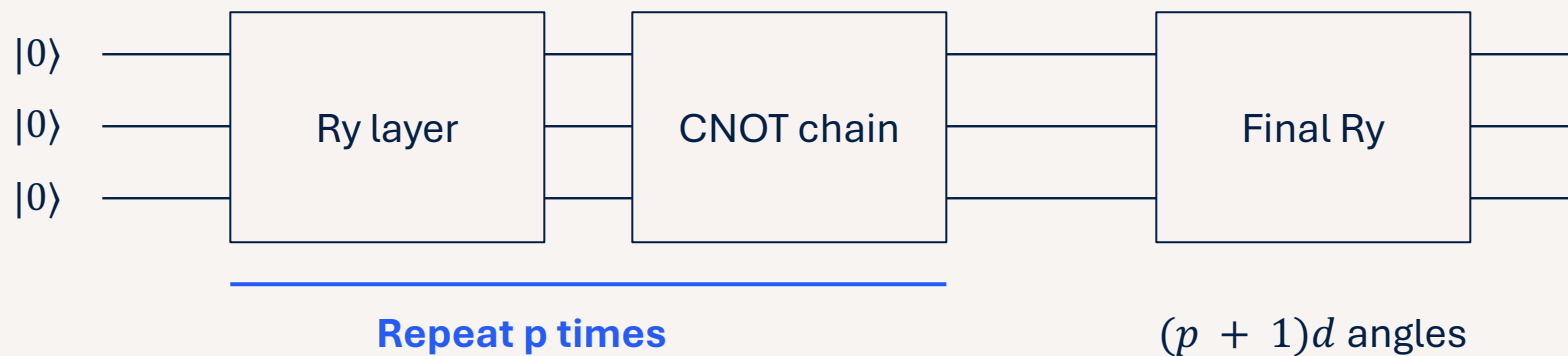
Expressivity
and optimization.

MAP recovery also needs a representable direction and successful optimization.

II. Proposed Variational Quantum Bayesian Regression

ENCODE A REAL DIRECTION IN A CIRCUIT

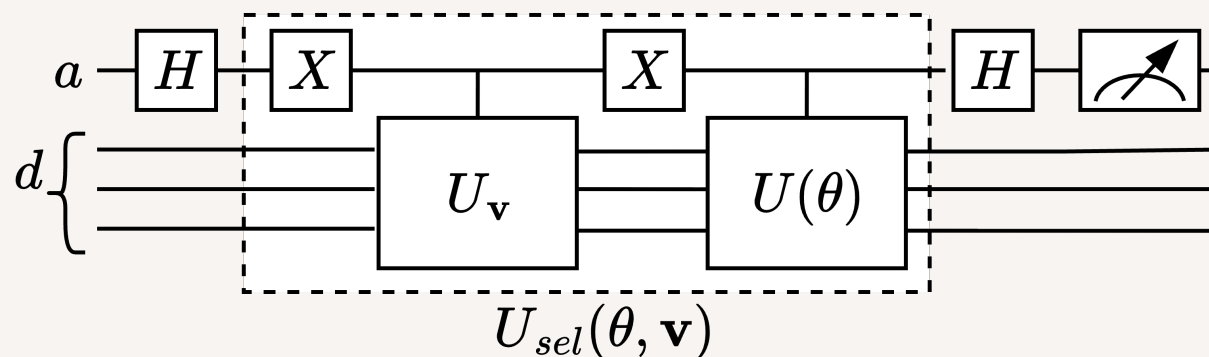
$$|\phi(\theta)\rangle = U(\theta)|0\rangle^{\otimes d}, \quad d = \lceil \log_2 D \rceil.$$



$D = 8, p = 2$: three data qubits, nine angles, and one ancilla for overlaps.

II. Proposed Variational Quantum Bayesian Regression

MEASURE OVERLAPS WITH ONE ANCILLA



$$s_i = \langle x_i / \|x_i\|_2 | \phi(\theta) \rangle = 2P_i(0) - 1$$

Retain each row norm classically to recover the unnormalized inner product.

II. Proposed Variational Quantum Bayesian Regression

ASSEMBLE FOUR MEASURED QUANTITIES

Data terms

$$g = \sum_i (\|x_i\|_2 s_i)^2.$$

$$c = \sum_i y_i \|x_i\|_2 s_i.$$

Prior terms

$$M = \sum_{\ell} \gamma_{\ell} P_{\ell}, \quad \mathbf{c} = M \mathbf{m}_0$$

$$d(\theta) = \sum_{\ell=1}^L \gamma_{\ell} \langle P_{\ell} \rangle.$$

$$e = \|\mathbf{c}\|_2 \langle \mathbf{c} / \|\mathbf{c}\|_2 | \phi \rangle.$$

$$\alpha = g + d, \quad \beta = c + e.$$

Combine data and prior terms; omit the prior overlap when its vector is zero.

II. Proposed Variational Quantum Bayesian Regression

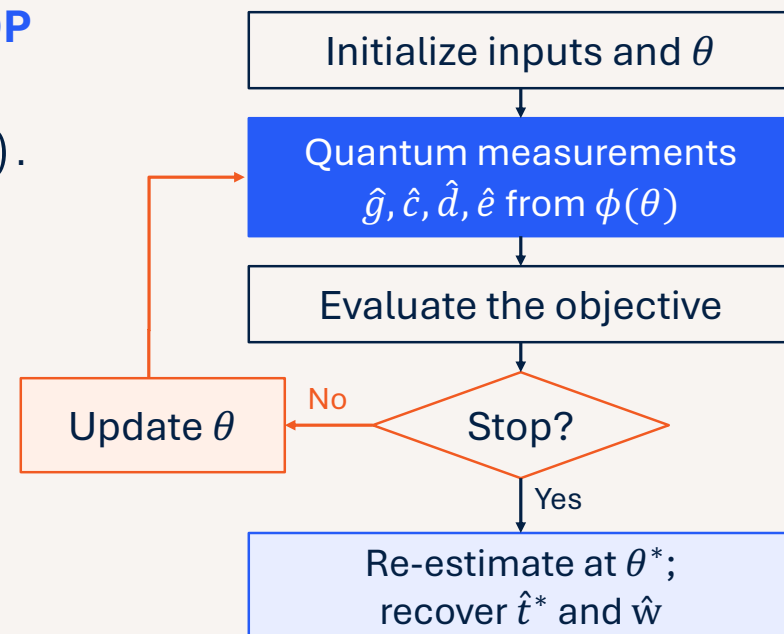
TRAIN WITH A HYBRID FEEDBACK LOOP

$$\hat{J}_{log} = \log (\hat{g} + \hat{d} + \delta) - \log ((\hat{c} + \hat{e})^2 + \delta).$$

$$\hat{t}^* = (\hat{c} + \hat{e}) / (\hat{g} + \hat{d}).$$

$$\hat{w} = \hat{t}^* \phi(\theta^*).$$

After optimization, re-estimate the components and recover the scale.



II. Proposed Variational Quantum Bayesian Regression

COMPUTATIONAL COST

Classical dense direct MAP:

$$\mathcal{O}(ND^2 + D^3) \text{ arithmetic}$$

N : training samples,

D : features,

p : ansatz layers,

L : prior Pauli terms,

S : shots per circuit,

T : Number of objective evaluations

Quantum register

$$d = \lceil \log_2 D \rceil$$

Trainable circuit params

$$(p + 1)d$$

Generic loading

$$\mathcal{O}(D) \text{ gates}$$

One objective evaluation

Total measurement budget

$$(N + L + 1)S \text{ shots}$$

$$\mathcal{O}(T(N + L + 1)S) \text{ circuit executions}$$

Avoids Gram construction; no demonstrated end-to-end speedup.

M. Möttönen, J. J. Vartiainen, V. Bergholm, and M. M. Salomaa, "Transformation of quantum states using uniformly controlled rotations," Quantum Inf. Comput., vol. 5, no. 6, pp. 467–473, Sep. 2005.

III. Experimental Results

SETTINGS

Synthetic

200 samples
 $D = 8, 16, 32$.

Energy Efficiency

768 samples
 $D = 8$; heating load.

Reference

SVD-based
direct MAP.

20% prior / 60% train / 20% test

Cosine similarity • Test RMSE • Test R^2

Simulation: mean $\pm$ SD over 20 seeds. Hardware: one separate run.

III. Experimental Results

SYNTHETIC REGRESSION: $N = 200, D = 8$

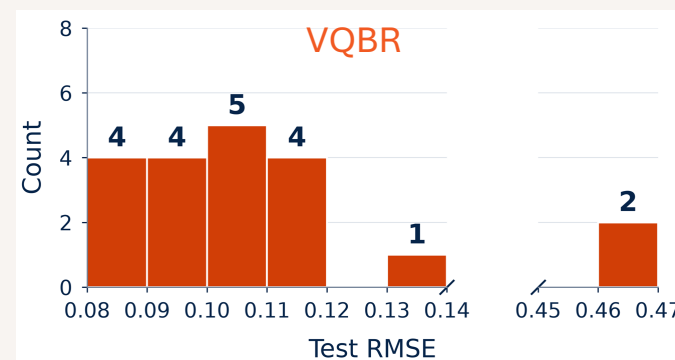
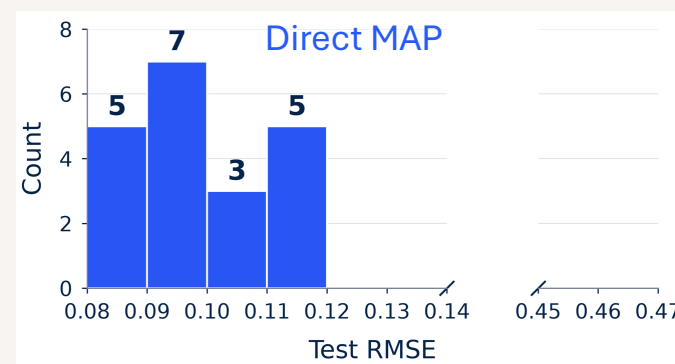
0.998 ± 0.005

Cosine similarity to the direct MAP direction

Default ansatz: $p = 2$

Mean $\pm$ SD over 20 seeds

Direction recovery is strong; VQBR test error varies more across seeds.



III. Experimental Results

ENERGY REGRESSION: $N = 768, D = 8$

0.927 ± 0.039

Direction cosine

Mean $\pm$ SD over 20 seeds

Test metric

Direct MAP

VQBR

$RMSE \downarrow$

2.943 ± 0.162

2.980 ± 0.173

$R^2 \uparrow$

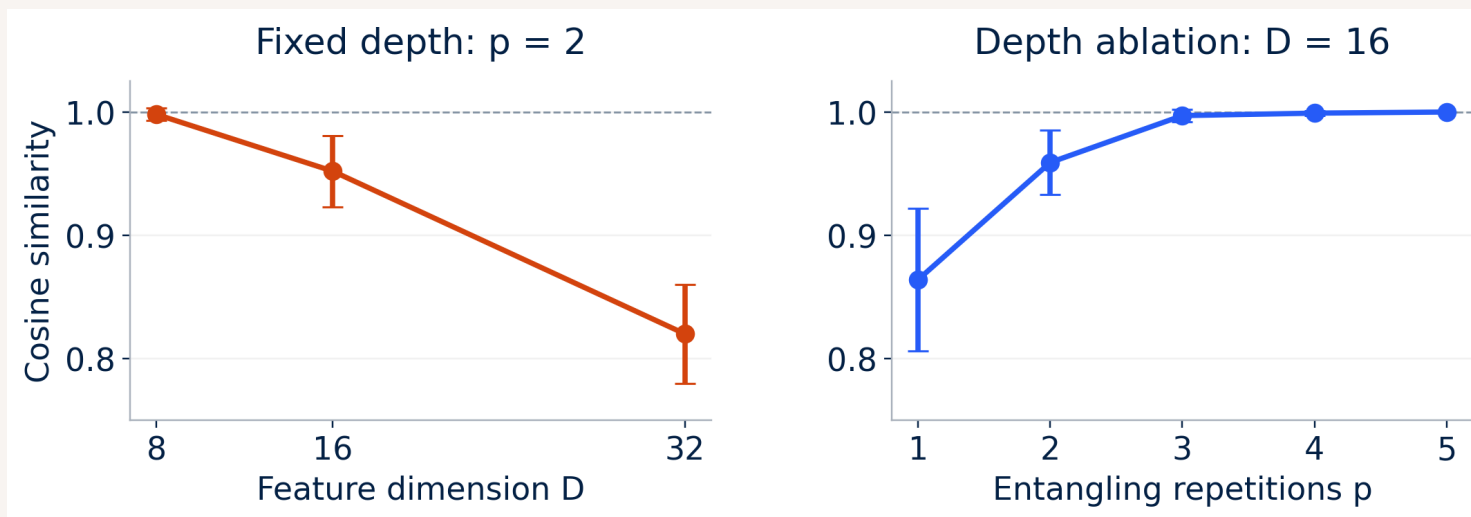
0.914 ± 0.011

0.912 ± 0.012

Predictive metrics are close even with less-than-perfect direction alignment.

III. Experimental Results

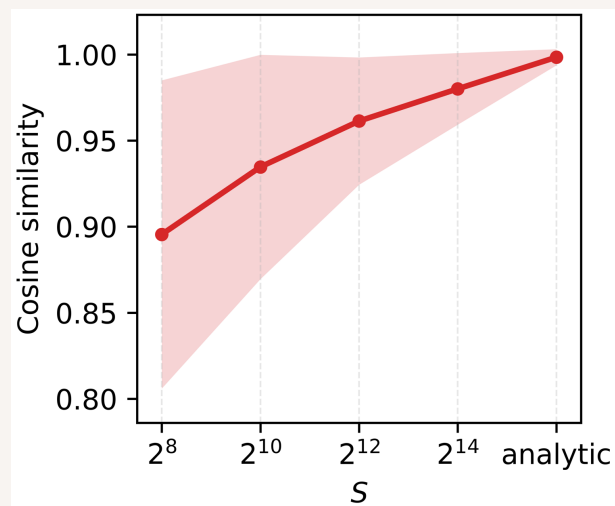
DIMENSION AND DEPTH ABLATIONS



At $D = 16, p = 5$ reaches 0.9999 ± 0.0001 cosine similarity.

III. Experimental Results

FINITE-SHOT DIRECTION RECOVERY



256 shots

0.895 ± 0.090

16,384 shots

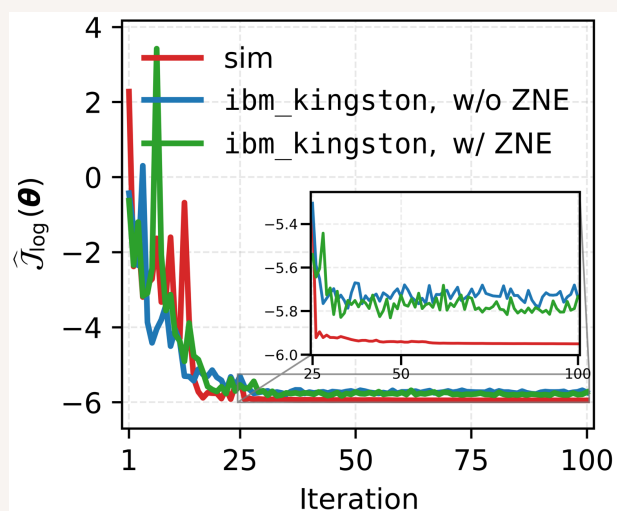
0.980 ± 0.021

Analytic: 0.998 ± 0.005

At $D = 8$, more shots improve recovery; mean $\pm$ SD over 20 seeds.

III. Experimental Results

IBM_KINGSTON HARDWARE DEMONSTRATION



Cosine similarity

With mitigation

0.9478

State vector reference

0.9956

One run: $N = 50$, $D = 8$, and $S = 1,024$; feasibility with error mitigation.

IV. Conclusions and Future Work

RESOURCE COSTS AND LIMITATIONS

Execution costs

$(N + L + 1)S$ shots per objective

Generic state loading: $O(D)$ gates

Optimization and output readout

L: Pauli terms; S: shots per circuit

Current scope

MAP point estimate

Restricted ansatz; small dimensions

One hardware run

No demonstrated runtime advantage

Compact qubit storage does not establish an end-to-end speedup.

IV. Conclusions and Future Work

Main contributions

- ❑ Exact scale elimination

Optimize directions within the chosen ansatz family.

- ❑ A measurement-based MAP objective

Avoids Gram construction, with initial simulation and hardware validation.

Next steps

Matched classical benchmarks, larger datasets, repeated hardware tests, and extensions toward posterior or predictive uncertainty.

References

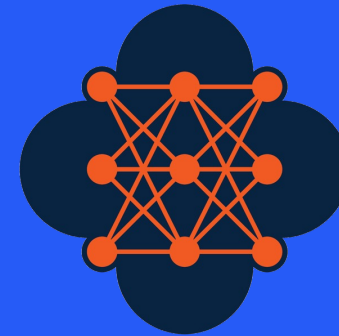
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